

CSI 4900 project plan: building on HiL-TV

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Background

When testing an image model, generated test images such as darker, foggier or compressed copies are sometimes no longer valid, because the object is too distorted to recognize. HiL-TV (Ghobari et al., ICSE-SEIP 2025) sorts valid from invalid image pairs using 13 image-similarity measures fed to a random forest, and a loop in which the model accepts the pairs it is confident about and a person labels a small batch each round. It reached about 97% accuracy and beat single-measure approaches by 4.6 to 15 points. Two things are left open: the similarity measures do not include newer image models, and the batch a person labels is picked at random.

Objectives

1. Repeat the HiL-TV experiments on the public CIFAR-10 car images from the replication package (1,574 image pairs; brightness, frost, contrast and JPEG changes) and check the results against the paper.
2. Addition A: add similarity measures from newer image models such as CLIP, and measure whether they improve accuracy for the same labelling effort.
3. Addition B: have the person label the images the model is least sure about instead of a random batch, and measure whether the same accuracy needs less human labelling.

Method

Data: the CIFAR-10 image pairs shipped in the paper's replication package, with the crowd-sourced valid or invalid labels from Hu et al. (ICSE 2022). The industrial images from the paper are not public, so only the CIFAR-10 half is repeated.

Repeating the paper: the 13 similarity measures as inputs, a random forest, and the labelling loop with the paper's confidence thresholds and batch sizes (fewer settings if time is short), 20 runs per setting, reporting accuracy, precision and recall against the share of images a person had to label.

The additions: Addition A computes the new measures once per image pair with pretrained models such as CLIP and DINOv2 on a free GPU and adds them as inputs. Addition B changes only how the batch is chosen, so the comparison with the paper is exact. One addition is completed, chosen with the supervisor; the other is the fallback.

Milestones

Week of	Milestone
Sep 22	Replication package running; CIFAR-10 image pairs and 13 measures recomputed
Oct 6	Accuracy-versus-labelling results repeated on CIFAR-10; compared with the paper
Oct 20	Comparison with the single-measure approaches (VIF and VAE) repeated
Oct 26 to 30	Reading week: get ahead or catch up
Nov 10	Addition implemented and running
Nov 24	Addition evaluated over 20 runs; accuracy-versus-labelling curves drawn
Dec 5	Report draft to supervisor; slides started
Dec 10 to 22	Presentation slot; final report submitted by the deadline

Deliverables

A cleaned, documented copy of the replication package with a requirements file, the repeated results, the addition's accuracy-versus-labelling curves, the written report, and the presentation.